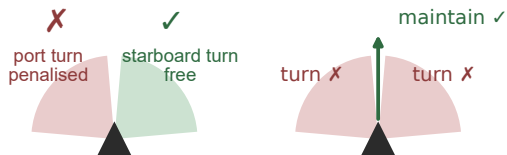


(a)

## Directional non-compliance penalty

Head-On · Give-Way · Overtaking

Stand-On

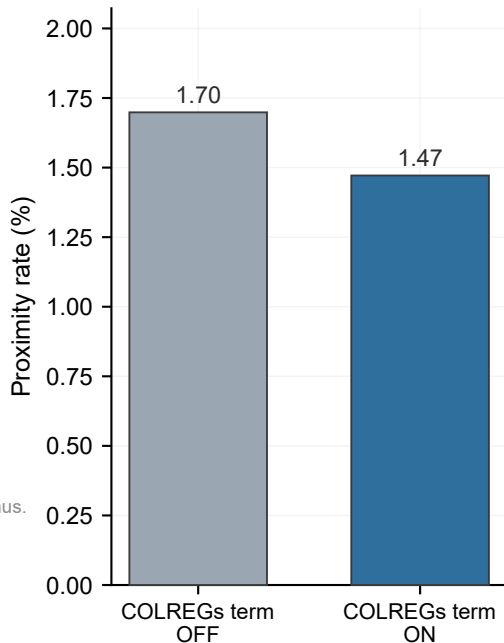


$$R_i^{\text{COLREGs}} = -\alpha_c (1 + \rho_i) \mathbf{1}[\rho_i > 0.3] (\dots)$$

A directional penalty on non-compliant turning — not a positive compliance bonus.  
Weighted by collision risk  $\rho_i$  and applied only above a risk gate.

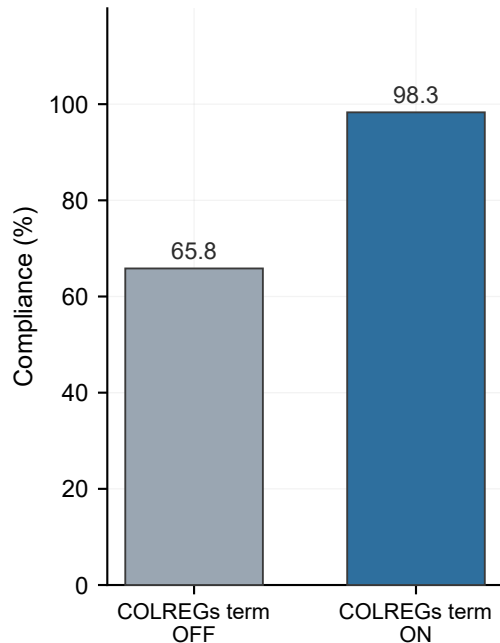
(b)

## Proximity rate



(c)

## Overall COLREGs (PRIMARY v2)



v12mix, 3 seeds. The COLREGs term raises PRIMARY v2 C and is the only ablation that moves C off the ceiling.  
Arrival and proximity stay similar; use C for this axis, not proximity alone.